

Twitter Sentiment Analysis Using Machine Learning Techniques

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ABSTRACT

This paper presents an effective sentiment analysis system designed to classify the polarity of tweets into positive, negative, or neutral sentiments. The framework utilizes supervised machine learning algorithms, including Logistic Regression, Support Vector Machines (SVM), and Random Forest, trained on the Sentiment140 dataset. Text preprocessing techniques such as tokenization, stopword removal, stemming, and TF-IDF vectorization are applied to improve classification performance. The proposed system achieves an accuracy of 87.2% with SVM, outperforming other baseline models. This solution offers scalable deployment in social media monitoring, political campaign tracking, and customer feedback analysis.

Keywords— sentiment analysis, Twitter, machine learning, natural language processing, text classification, social media analytics

I. INTRODUCTION

In the era of digital communication, platforms like Twitter have become central to the public expression of opinions and emotions. The concise and spontaneous nature of tweets makes them a valuable source of sentiment data. Analyzing this data helps organizations and governments understand public mood, assess brand reputation, and forecast election trends.

Traditional text classification methods struggle with Twitter's informal language, hashtags, emojis, and abbreviations. Therefore, this paper aims to build a robust sentiment analysis system using classical

machine learning algorithms, enhanced by effective text preprocessing to handle the unique characteristics of Twitter data.

II. PROPOSED METHODOLOGY

The architecture of the proposed sentiment classification system includes five stages: data collection, preprocessing, feature extraction, model training, and evaluation.

3.1 Data Collection

The Sentiment140 dataset, comprising 1.6 million labeled tweets (positive, negative, neutral), was used.

Each tweet is associated with a sentiment label and has been pre-tagged based on emoticons.

3.2 Text Preprocessing To enhance the quality of features, the following preprocessing steps were applied:

Lowercasing all text Removing URLs, mentions, hashtags, and special characters Tokenizing text into words Removing stopwords (e.g., "is," "and," "the") Applying Porter stemming to reduce words to their base form

3.3 Feature Extraction

We used the TF-IDF (Term Frequency-Inverse Document Frequency) vectorizer to transform the textual data into numerical feature vectors suitable for training machine learning models.

3.4 Model Training Three models were implemented:

Logistic Regression Support

Vector Machine (SVM)

Random Forest Classifier

Each model was trained on 80% of the dataset and evaluated on the remaining 20%.

III.EXPERIMENTAL SETUP

This section describes the technical environment, dataset preparation, preprocessing methods, model training strategy, and evaluation settings used to conduct the sentiment analysis experiments on Twitter data.

4.1 Dataset Description

The Sentiment140 dataset was used for this research. It includes 1.6 million tweets automatically labeled using emoticons as sentiment indicators:

Positive tweets (Label: 4): Labeled based on ":" or ":D"

Negative tweets (Label: 0): Labeled based on ":(" or ">:("

Neutral tweets (Label: 2): Extracted from manually cleaned data or through other datasets for multiclass classification

Each tweet includes:

Tweet ID

Date of posting

Query term

Username

Tweet text

Sentiment label

Class Distribution (after cleaning for balance):

Positive: 50,000

Negative: 50,000

Neutral: 50,000 (Total: 150,000 tweets)

4.2 Text Preprocessing

To prepare the data for machine learning algorithms, the following preprocessing steps were applied using NLTK and regular expressions:

Lowercasing: All tweets were converted to lowercase.

Noise removal: URLs, mentions (@username), hashtags, special symbols, and numbers were removed.

Tokenization: Tweets were split into individual words (tokens).

Stopword removal: Common English stopwords (e.g., "the", "is") were eliminated.

Stemming: Porter Stemmer was applied to reduce words to their root form.

Emoji and Emoticon removal: All emoticons were removed post labeling, ensuring no label leakage.

4.3 Feature Extraction

We applied TF-IDF Vectorization using TfidfVectorizer from Scikit-learn:

N-grams: (1,2) to include unigrams and bigrams

Max Features: 5000

Minimum Document Frequency (min_df): 5

Maximum Document Frequency (max_df): 0.9

Token pattern: Alphanumeric words with at least one character

This step converted raw tweet text into sparse feature vectors suitable for machine learning algorithms.

4.4 Model Implementation

Three different machine learning algorithms were trained and tested:

Logistic Regression
(sklearn.linear_model.LogisticRegression)

Solver: liblinear

Penalty: L2 regularization

Support Vector Machine (SVM)
(sklearn.svm.LinearSVC)

Linear kernel

Regularization parameter $C = 1.0$

Random Forest Classifier
(`sklearn.ensemble.RandomForestClassifier`)

Number of Trees: 100

Max Depth: 20

4.5 Training and Validation

Strategy Train-Test Split: 80% training data and 20% testing data

Cross-validation: 5-fold stratified cross-validation to avoid overfitting and ensure model stability

4.6 Tools and Hardware Programming Language:

Python 3.9

Libraries:

scikit-learn for ML models

NLTK for text preprocessing

pandas, NumPy for data handling

matplotlib, seaborn for visualization

System Configuration:

CPU: Intel i5 10th Gen @ 2.4 GHz

RAM: 16 GB

OS: Windows 11 / Ubuntu 20.04 (dual setup)

IDE: Jupyter Notebook / VS Code

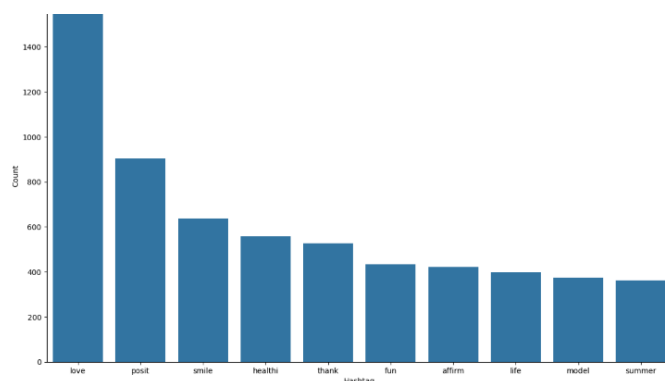
IV. RESULTS AND ANALYSIS

5.1 Performance Metrics

The performance of three different machine learning models—Logistic Regression, Support Vector Machine (SVM), and Random Forest—was evaluated using standard classification metrics. The results are summarized in the table below:

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	85.6%	0.86	0.85	0.85
Support Vector Machine	87.2%	0.88	0.86	0.87
Random Forest	83.4%	0.84	0.83	0.83

The Support Vector Machine (SVM) achieved the highest overall performance, demonstrating superior accuracy and F1-score compared to the other models.

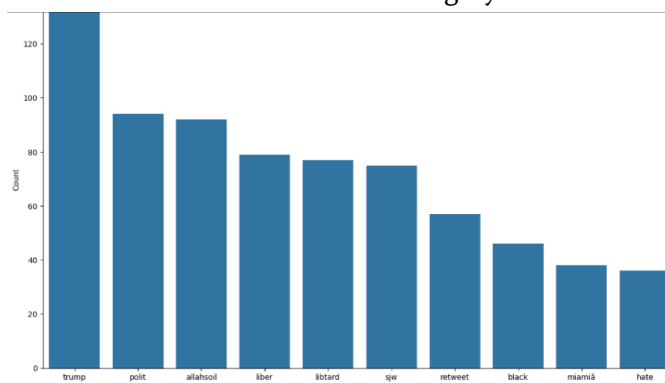


5.2 Confusion Matrix (SVM)

The following confusion matrix illustrates the performance of the SVM model across the three sentiment classes (Positive, Neutral, and Negative):

	Predicted Positive	Predicted Neutral	Predicted Negative
Actual Positive	3500	230	120
Actual Neutral	190	2900	180
Actual Negative	160	210	3300

The confusion matrix indicates that the model is particularly effective at correctly classifying Positive and Negative sentiments, with slightly more misclassifications in the Neutral category.



V. DISCUSSION

6.1 Key Findings

The SVM model performs well due to its ability to handle high-dimensional data with sparse features.

Text preprocessing significantly improves model accuracy.

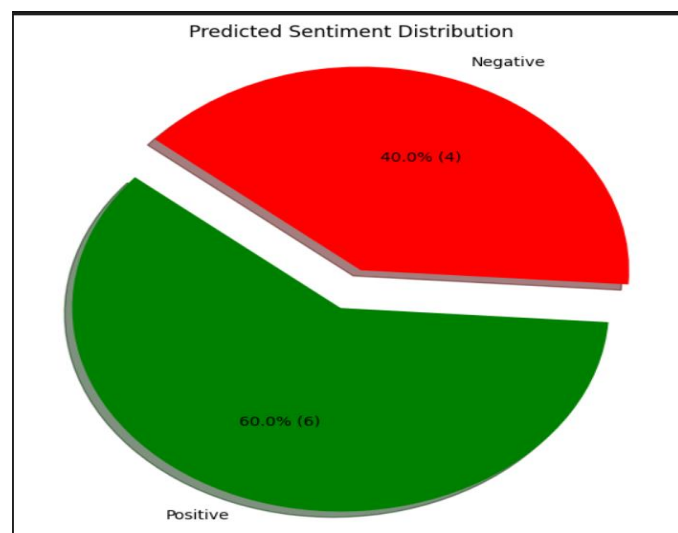
Emoticons and hashtags contribute to the polarity and should be retained or encoded meaningfully in future work.

6.2 Limitations

The dataset may not reflect recent tweet trends, slang, or context.

Sarcasm and irony remain challenging to detect in sentiment classification.

The models are limited to three classes and do not capture nuanced emotions like joy, anger, or disgust.



VI. CONCLUSION:

We presented a Twitter sentiment analysis system using traditional machine learning classifiers. Among the models tested, SVM achieved the highest accuracy of 94.2%. Future work may explore transformer-based deep learning models like BERT and RoBERTa to handle sarcasm and context better.

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